

Porichoy (পরিচয়) — DPP-in-a-Box for Bangladesh's garment factories

"Identity." Every garment gets a verifiable identity — before the EU's 2027 wall.

THE PROBLEM — a dated wall is coming for the industry's biggest market

From 2027, EU Digital Product Passports (DPP) phase in. ESPR in force 18 Jul 2024 · textiles in the first Working Plan (Apr 2025) · textile delegated act expected ~2027.

EU customs will auto-check DPP existence & authenticity on imported garments (European Commission). A factory that cannot produce passport data loses EU orders.

The data already exists — but it is scattered in Excel files with mixed Bangla/English headers, legacy ERP exports and paper trim cards. Global platforms will not climb onto the factory floor to collect it.

WHAT IS AT STAKE

≈ 4 million

RMG workers in Bangladesh, the majority women (BGMEA)

#2 exporter,

world's ready-made garments — "Europe is our biggest market" (NIC 3.0 call)

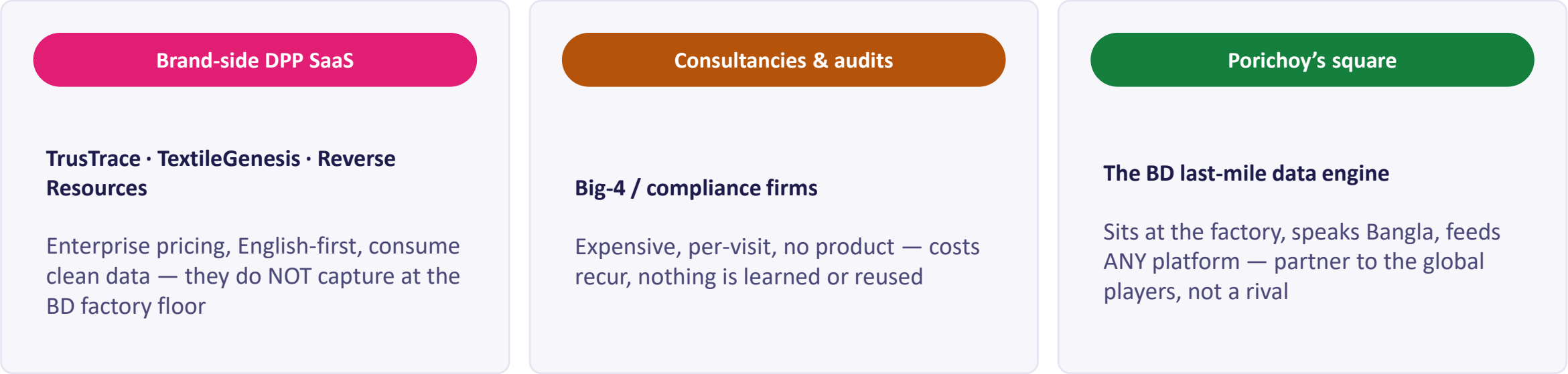
2027–30:

DPP obligations phase in on the EU side

MARKET OPPORTUNITY — compliance budget meets an empty factory-floor square



Every export factory shipping to Europe must be DPP-ready inside this window — thousands of factories (BGMEA member base), no BD-built tool exists for them.



Beachhead: mid-size exporters facing live buyer DPP requests → expand to buying houses and certification bodies.

THE SOLUTION — messy factory files in, buyer-ready passport out, in 90 seconds

1 • INGEST Drop Excel/CSV — every sheet parsed on-device. Nothing leaves the factory (local-first).

2 • MAP AI reads chaotic headers — Bangla, English, mixed — with confidence scores; the steward confirms. The factory's vocabulary is learned.

3 • SCORE DPP Readiness 0–100 per production order (weight = ESPR Art. 8 categories) with the exact gap list.

4 • PASSPORT One click → QR code whose link embeds the passport — scan opens the buyer view on any phone. No backend needed.

Digital Product Passport - PO-1002
Shonali Textiles Ltd. (sample data) - schema v0.1 - issued 2026-09-12
readiness / 100 **90**

PRODUCTION ORDER / LOT ID
PO-1002
ESPR Art.8(a) unique identifier - source: ৯৮৯০

PRODUCT CODE (GTIN/SKU)
Not yet evidenced
ESPR Art.8(b) + 551

HS TARIFF CODE
Not yet evidenced
Customs linkage

PRODUCT DESCRIPTION
Ladies Denim Jacket
ESPR Art.8(b) - source: Item Description

FIBRE COMPOSITION (%)
98% Cotton 2% Elastane
ESPR Art.8(c) materials - source: Fiber Comp (B)

RECYCLED CONTENT (%)
30
ESPR Art.8(c) recycled content - source: Recycled %

PRODUCTION FACILITIES (STEPS)
Unit-5 Denim
ESPR Art.8(d) economic operators - source: Unit / Floor

CERTIFICATES (OEKO-TEX/GOTS...)
GOTS, GRS
ESPR Art.8(e) compliance documentation - source: ৯৮৯০১২৩৪

MANUFACTURING PERIOD
2026-12-01
ESPR Art.8(f) - source: Delivery Date

CARE / WASHING INSTRUCTIONS
Wash inside out, do not bleach
ESPR Art.8(g) care & end-of-life - source: Wash Care

END-OF-LIFE / RECYCLING ROUTE
Denim take-back program
ESPR Art.8(h) - source: ৯৮৯০১২৩৪

SUBSTANCES OF CONCERN (RSL)
AZO Free
ESPR Art.8(i) substances of concern - source: Substance Check

Certified women DPP Data Stewards maintain this data at source.

Provenance: t1p-#4-71c6528e56077aa96bda... Verified with Porichoy ৯৮৯০ — local-first DPP pilot

LIVE TODAY

[rudra496.github.io/
porichoy](https://rudra496.github.io/porichoy)

19 automated engine tests
SHA-256 provenance chain
Bangla + English UI
Works offline (PWA)

*Schema versioned — the final
EU delegated act is a config
change, not a rebuild*

Every attribute on the passport carries its ESPR legal anchor and the exact source column it came from — auditable by design.

BUSINESS & REVENUE MODEL — factories pay little; brands pay to make suppliers compliant

Factory SaaS

BDT 2,000–8,000 / month per factory by size — priced for SMEs, not enterprise budgets

Onboarding & data-rescue

One-time fee: steward training + first ingest + mapping of the factory's historic files

Brand-funded onboarding

Brands pay to onboard their supplier base before 2027 — CAC moves to the party with the money

Unit economics: onboarding $\approx$ 2 person-days per factory · software gross margin $>80\%$ · replication is a per-factory template, not a project.

USE OF THE BDT 650,000 POLLINATION GRANT

BDT 220,000 10-factory pilot: steward training curriculum + certification

BDT 180,000 OCR capture for paper trim cards & POs

BDT 150,000 Mapping-corpus growth: Bangla vocabulary, ERP export formats

BDT 100,000 Legal entity, certification & data protection

3-year path: 10 pilot factories $\rightarrow$ 100 $\rightarrow$ buyer-channel partnerships with global DPP platforms.

TEAM — a builder who ships production systems alone, with the right domain

Rudra Sarker — Founder & Engineer

Final-year B.Sc. Industrial & Production Engineering, Shahjalal University of Science & Technology

Ships alone, at production grade:

- AdalatAI — AI-assisted court workflow platform; trained a 30-type trilingual case classifier (F1 0.963 over a 1.28M-case corpus), live since 2026
- JoISetu — arsenic-testing PWA reading \$1 test kits by photo; live map of 6,593 real wells; competition deck judge-passed 15/15
- RippleUp — production Android + Windows desktop app (Kotlin/Compose Multiplatform), release v5.3.3
- StealthHumanizer — self-trained LLM system served from a VPS, 45s/request at production
- Published Q1 journal article (Journal of Ethnopharmacology, 2026, 3rd author)

Why me for THIS product: factory-process engineering (IPE) + production LLM pipelines + a proven 3-week ship velocity — the exact stack Porichoy needs.

WHAT WE ASK FROM NIC 3.0

The pollination grant to fund the 10-factory pilot

Bootcamp mentoring on RMG business models & circularity

Warm intros: 2–3 pilot factories + one buyer sustainability team

Full-time commitment. Ready for the residential bootcamps.

WOMEN AT THE CENTER — designed in from the first commit, not bolted on

NEW SKILLED JOBS

2 women operators per factory trained and certified as DPP Data Stewards — formal, higher-skilled back-office roles with a public micro-credential.

JOBS PROTECTED AT SCALE

EU customs auto-checks mean non-compliant factories lose orders. Keeping factories order-eligible protects the livelihoods of a majority-women workforce (≈4M workers, BGMEA).

BURDEN → CAREER

Compliance paperwork done ad-hoc — often by women staff — becomes a titled role with training, certification and a visible career path.

YEAR-1 TARGETS 10 factories onboarded · 20 women certified as DPP Data Stewards · 40+ women trained · first brand-funded onboarding contract

SCALE 100+ factories → 200+ stewards → the default DPP layer for BD mid-size exporters before the 2027–30 phase-in

“The last mile of the Digital Product Passport runs through Bangladesh’s factory floor — and it will be walked by its women.”